

Research Interests

- Developing climate datasets and methods, including leading the DCENT(-I) temperature datasets (Refs. 25&36 below)
- Understanding climate variability and dynamics, and improving long-term projections using simplified models and AI
- Assessing climate risks in collaboration with policy-relevant and practitioner-led research

Education

2015-21 Ph.D. in Earth and Planetary Sciences, Harvard University (Advisor: Peter Huybers)
2013-15 M.S. in Meteorology, Nanjing University, China
2009-13 B.S. in Applied Meteorology and Minor in Finance, Nanjing University, China

Appointments

2023- Lecturer, School of Ocean and Earth Science (SOES), University of Southampton (UoS)
2021-23 Postdoctoral Scholar, Physical Oceanography Department, WHOI

Awards & Honours

Fellowship Future Leaders Fellow (2025-, UKRI, £1.41M) | CHEP Education Fellow (2025-, UoS) | Weston Howland Jr. Postdoctoral Fellow (2021, WHOI) | High Meadows Environmental Fellow (2021, Princeton, Declined) | Harvard Horizons Fellow (2020, Harvard) | William Benjamin and Jill Kowal Graduate Aid Fund in Environmental Studies (2015-16, Harvard)

Keynote Speaker 2nd Tropical Climate Variability Conference (2024, Bremen)

Plenary Speaker 15th International Meeting on Statistical Climatology (2024, Toulouse)

Teaching & Supervision (*† incoming students; * co-supervised*)

Lecturer Computational Data Analysis for Geophysicists and Ocean Scientists (Fall, 2023-)

Ph.D. Supervisor Callum Pemberton-Louden (2025-) | Ewan Strathdee (2025-) | Christian Kenwright(2025-) | Yifei Fan*(2021-)

Ph.D. Mentor Se-Yong Song (2023-) | Chenggong Wang (2021-22) | Glenn Liu (2021-24)

Undergraduate Mentor Charlotte Henke (2021) | Sarah King (2020-2021)

Professional Services

UoS Committee EDI officer, UoS Math For Our Future Climate Centre for Doctoral Training (2024-) | Academic Responsibility and Conduct Officer, SOES (2024-) | Member, SOES Oceanography Programme Re-branding Committee (2025-) | Member, UoS MOAP bid working group (2025-) | Member, SOES Climate Hiring Committee (2024)

Guest Editor Atmosphere Special Issue (Volume 15, Issue 12)

Reviewer PNAS | Nature Communications | Science Advances | Quarterly J. Royal Meteorological Society | J. Climate | Geophysical Research Letter | JGR: Atmospheres | Earth's Future | npj Climate and Atmospheric Science | Climate Dynamics | J. Atmospheric and Oceanic Technology | NOAA Small Business Research Funding

Presentation Judge AGU (2022) | Ocean Science Meeting (2022) | National Collegiate Research Conference (2022)

Organizer AGU co-convener (2023; GC084) | Harvard ClimaTea seminar (2017)

Selected Media Coverage & Outreach

Media Agence France-Presse (08/09/2025) | AGU EOS (08/09/2025; 05/02/2024; 20/10/2023) | Associated Press (08/09/2025) | CNN (08/09/2025) | Financial Times (01/03/2026) | Guardian (15/09/2025) | National Public Radio (10/09/2025; 19/08/2019) | Nature News & Views (20/11/2024; 17/07/2019) | New Scientist (01/06/2025; 10/02/2025; 10/01/2025) | Science Insider (31/01/2024) | Scientific American (08/09/2025) | UK's Science Media Centre (05/02/2024) | Yale Climate Connections (07/01/2016)

Outreach Voluntary teaching at Perry School (public middle school in south Boston; Winter, 2019-20)

Peer-reviewed Publications (* co-first author; † student or mentee; — corresponding author)

- [36] **Chan D.**, Chan S. C., Siddons J. T., Cable A., Faulkner A., Cornes R. C., Kent E. C., Gebbie G., & Huybers P. (2026). DCENT-I: A Globally Infilled Extension of the Dynamically Consistent ENsemble of Temperature Dataset. *Geoscience Data Journal*, 13(2), e70054.
- [35] Cornes R. C., Chan S. C., Cable A., **Chan D.**, Faulkner A., Kent E. C., & Siddons J. T. (2026). GloMarGridding: A Python Toolkit for Flexible Spatial Interpolation in Climate Applications. *Geoscience Data Journal*, 13(2), e70064.
- [34] He P., Xu Z.*, **Chan D.**, Liu P., & Bai Y. (2025). Climate change aggravates the over-consumption of added sugar in the U.S. especially in disadvantaged groups. *Nature Climate Change* 15, 963–970.
- [33] Proctor, J., Vargas Z., L., **Chan, D.**, Huybers, P. (2025). Climate change increases the interannual variance of summer crop yields globally through changes in temperature and water supply. *Science Advances*, 11(36), eady3575.
- [32] Yin, X., Schlegel, R. W., Huang, B., **Chan D.**, Graham, G., Hu, Z.-Z., Zhang, H.-M. (2025). Sea Surface Temperature. In "State of the Climate in 2024". *Bulletin of the American Meteorological Society*, 106(8), S180–S185.
- [31] Liao, W., Wang, L., Liu, X., **Chan D.**, Li, D. (2025). Standardized heat islands and persistence drive modeled urban heat events. *Nature Cities*, 1-8.
- [30] **Chan D.**, Gebbie G., & Huybers P. (2025). Re-evaluating Historical Sea Surface Temperature Datasets: Insights from the Diurnal Cycle, Coral Proxy Data, and Radiative Forcing. *GRL*, 52(13), e2025GL116615.
- [29] **Fan Y.†**, **Chan D.**, Eugene E., Zhang P., & Li L. (2025). Subpolar North Atlantic cooling reinforced by colder, drier atmosphere with a weakening Atlantic Meridional Overturning Circulation. *Science Advances*, 11(23), eads1624.
- [28] Kent E.C., **Chan D.**, Cornes R., Englyst P., et al. (2025). Global surface temperatures. In *Energy and Climate Change* (pp. 85-120). Academic Press.
- [27] **Wang C.†**, Yang W., Vecchi G., Zhang B., Soden B. J., & **Chan D.** (2025). Diagnosing the factors that contribute to the intermodel spread of climate feedback in CMIP6. *Journal of Climate*, 38(3), 663-674.
- [26] Sippel S., Kent E.C., Meinshausen N., **Chan D.**, et al. (2024). Early-twentieth-century cold bias in ocean surface temperature observations. *Nature*, 635, 618–624.
- [25] **Chan D.**, Gebbie G., Huybers P., & Kent E. (2024). A Dynamically Consistent ENsemble of Temperature at the Earth surface since 1850 from the DCENT dataset. *Scientific Data*, 11(1), 953.
- [24] **Fan Y.†**, **Chan D.**, Zhang P., & Li L. (2024). Disagreement on the North Atlantic Cold Blob Formation Mechanisms among Climate Models. *Journal of Climate*, 37(16), 4061-4078.
- [23] **Chan D.**, Gebbie G., & Huybers P. (2024). Ensemble of land-surface air temperatures between 1880-2022 using revised pair-wise homogenization algorithms accounting for auto-correlation. *Journal of Climate*, 37(7), 2325–2345.
- [22] Zhang W., Yao Y., **Chan D.**, & Feng J. (2024). Advances in Air–Sea Interactions, Climate Variability, and Predictability (Editorial). *Atmosphere*, 15(12), 1422.
- [21] Yin X., Huang B., **Chan D.**, Graham G., Hu Z., Zhang H. (2024) Sea-surface temperatures [in "State of the Climate in 2023"]. *BAMS*, 105(8), S163–S168.
- [20] Ridgen A., Golden C., **Chan D.**, & Huybers P. (2024). Climate change linked to ongoing drought in Southern Madagascar. *npj Climate and Atmospheric Science*, 7(1), 41.
- [19] Bao X., Zhang S., Jiang G., **Chan D.**, Hu Y., Wu H., Li H., Wang X., & Yang T. (2023). Climate changes in the Cryogenian nonglacial epoch: A global synthesis with new findings from the Datangpo Formation in South China. *Global and Planetary Change*, 229, 104234.
- [18] Yin X., Huang B., Hu Z., **Chan D.**, Zhang H. (2023) Sea-surface temperatures [in "State of the Climate in 2022"]. *BAMS*, 104(9), S153–S156.
- [17] **Chan D.**, Gebbie G., & Huybers P. (2023). Global and Regional Discrepancies between Early 20th Century Coastal Air and Sea-Surface Temperature Detected by a Coupled Energy-Balance Analysis. *Journal of Climate*. 36(9), 2205-20.
- [16] Proctor J., Ridgen A., **Chan D.**, & Huybers P. (2022). Soil moisture measurements improve prediction of crop yields and reduce projected climate change damages. *Nature Food*, 3(9): 753.
- [15] **Chan D.**, Ridgen A., Proctor J., Chan P. H. & Huybers P. (2022). Differences in radiative forcing, not sensitivity, explain differences in summertime land temperature variance change between CMIP5 and CMIP6. *Earth's Future*,

e2021EF002402.

- [14] **Chan D.**, Vecchi G., Yang W. & Huybers P (2021). Improved simulation of 19th- and 20th-century North Atlantic hurricane frequency after correcting historical sea surface temperatures. *Science Advances*. 7(26), eabg6931.
- [13] **Chan D.**, & Huybers P (2021). Correcting sea surface temperature observations removes World War II warm anomaly. *Journal of Climate*, 34(11), 4585-602.
- [12] **Chan D.** (2021). Combining statistical, physical, and historical evidence to improve historical sea surface temperature records. *Harvard Data Science Review*. 3(1), doi: 10.1162/99608f92.edcee38f
- [11] **Dai C.**, **Chan D.***, Huybers P., & Pillai, N. (2021). Late 19th-century navigational uncertainties and their influence on sea surface temperature estimates. *Annals of Applied Statistics*, 15(1): 22-40.
- [10] **Chan D.**, & Huybers P. (2020). Systematic differences in bucket sea surface temperatures caused by misclassification of engine room intake measurements. *Journal of Climate*. 33(18), 7735–53
- [9] **Chan D.**, Cobb A., Vargas L., Battisti D., & Huybers P. (2020). Summertime temperature variability increases with local warming in mid-latitude regions. *Geophysical Research Letters*, e2020GL087624.
- [8] **Chan D.**, Zhang, Y., Wu Q., & Dai X. (2020). Quantifying the dynamics of the interannual variabilities of the wintertime East Asian Jet Core. *Climate Dynamics*, 54(3), 2447-63.
- [7] **Chan D.**, Kent E., Berry D. & Huybers P. (2019). Correcting datasets leads to more homogeneous early 20th century sea surface warming. *Nature*, 571, 393-397. (covered by [NPR](#))
- [6] **Chan D.** & Huybers P. (2019). Systematic differences in bucket sea surface temperature measurements amongst nations identified using a linear-mixed-effect method. *Journal of Climate*, 32(5), 2569-89.
- [5] **Chan D.**, **Wu Q.**, Jiang G., & Dai X. (2016). Projected shifts in Köppen climate zones over China and their temporal evolution in CMIP5 multi-model simulations. *Advances in Atmospheric Sciences*, 3(33), 283-93.
- [4] **Hu C.**, Wu Q., Yang S., Yao Y., **Chan D.**, Li Z., & Deng K. (2016). A linkage observed between austral autumn Antarctic Oscillation and preceding Southern Ocean SST anomalies. *Journal of Climate*, 29(6), 2109-22.
- [3] **Wu Q.**, Cheng L., **Chan D.**, Yao Y., Hu H., & Yao Y. (2016). Suppressed mid-latitude summer atmospheric warming by Arctic sea ice loss during 1979–2012. *Geophysical Research Letters*, 43(6), 2792-800.
- [2] **Chan D.**, & **Wu Q.** (2015). Significant anthropogenic-induced changes of climate classes since 1950. *Scientific Reports*. 5. 13487. (covered by [Yale Climate Connections](#))
- [1] **Chan D.**, & **Wu Q.** (2015). Attributing observed SST trends and sub-continental land warming to anthropogenic forcing during 1979–2005. *Journal of Climate*, 28, 3152–70.

Sea-Going Experience

2021 One Ocean Expedition, Statsraad Lehmkuhl: Miami-New York, Dec. 10-18

Conferences and Presentations

Invited Talks

- [7] Re-Evaluating Historical SST Data Sets Implications for proxy calibration and paleo-data assimilation (AGU 2025)
- [6] Recent advancements in historical earth surface temperature analysis and insights into climate change from enhanced data. (U East Anglia, 2025 | Pacific Climate Impacts Consortium, 2024 | Helmholtz Centre for Polar and Marine Research, 2024 | U Southampton (Math), 2024 | U Leipzig, 2024 | U Bremen, 2024 | UK Met Office, 2024 | Plenary Talk, IMSC, France, 2024 | UCL, London, 2024)
- [5] Combining the physics of air-sea interaction and data-driven methods to improve historical estimates of earth surface temperatures (Ocean University of China, 2023 | Duke Kunshan, 2023 | Hanyang University, 2023 | MIT, 2023 | UC Colorado, 2023 | NCAR, 2023 | U Chicago, 2023 | WHOI GFD summer school, 2022 | U Miami, 2022).
- [4] Are we already at a 1.5°C warming threshold? (U Southampton (SOES), 2022).
- [3] Combining statistical, physical, and historical methods to improve historical sea surface temperature data (Zhejiang University, 2022 | Ocean Dynamics Seminar, 2022 | Penn State U, 2022 | UC Irvine, 2021 | U Washington, 2021 | WHOI, 2021 | Nanjing University, 2021 | U.K. National Oceanography Centre, 2021 | Harvard Horizons, 2021 | Princeton, 2020 | Yale, 2020).
- [2] Applying statistical methods to climate reconstructions – Late 19th-century navigational errors and their influence on sea surface temperatures (Joint Statistical Meeting, 2020).

- [1] Correcting datasets leads to more homogeneous early-twentieth-century sea surface warming (Fudan University, 2019 | Nanjing University, 2019).

Recent Conference Talks

- [3] SST Dataset Choice Affects Estimates of Historical Climate Variability (EGU 2026)
- [2] Improved Homogenisation of Observation Shows Steadier and Faster Historical Global Warming (AMS, 2025).
- [1] Discrepancies between Coastal Air and Sea-Surface Temperature and Implications for Global Mean Temperature Estimates (AMS, 2023 | AGU, 2022 | 47 NOAA Climate Diagnostic and Prediction Workshop, 2022).